

THARUN KUMAR KATABATHINA

Data Analyst | Business Analyst | Solution Analyst

Chicago, IL | Open to Relocation | tharun312k@gmail.com | <https://github.com/Tharun-Kumar-Katabathina>
[linkedin.com/in/tharunkumarkatabathina](https://www.linkedin.com/in/tharunkumarkatabathina)

PROFESSIONAL SUMMARY

Versatile Computer Science graduate (MS, GPA 3.92) with comprehensive experience spanning data analytics, business intelligence, solution design, and enterprise systems development. Skilled in Python, SQL, R, Power BI, Tableau, SAS, SPSS, Excel financial modeling, and statistical analysis with applied domain knowledge in healthcare, pharmaceutical, financial, and retail analytics. Proven operational and compliance background through digital accessibility workflow management and cross-functional stakeholder collaboration.

TECHNICAL SKILLS

Analytics & BI: Python (Pandas, NumPy, Scikit-learn), SQL, R, Big Data Analytics, Power BI (DAX), Tableau, Excel Advanced Modeling

Statistical & Domain Tools: SAS, SPSS, STATA, A/B Testing, Regression, Time Series (ARIMA), Epidemiological Analysis, Market Research, Accounting Principles

Finance & Business: Financial Modeling, Budgeting, Forecasting, Variance Analysis, Revenue Analytics, Operational Analysis, SAP (Familiarity)

Solution & Systems Design: Requirements Gathering, Gap Analysis, Business Process Modeling, Technical Specifications, REST APIs, Microservices (Kafka, Docker, AWS)

Development: Python, Java, JavaScript, Node.js, React.js, Spring Boot, FastAPI, MySQL, MongoDB, PostgreSQL, Git/GitHub

Compliance & Process: WCAG 2.1 / ADA / Section 508, Agile/Scrum, SDLC, QA Testing, SLA Monitoring, Workflow Automation, JIRA

Soft Skills: Stakeholder Communication, Executive Presentation, Independent Problem-Solving, Organizational Discipline, Collaborative Teamwork, Active Learner

PROFESSIONAL EXPERIENCE

Digital Accessibility Remediation Analyst | *Office of Digital Accessibility Univ. of Illinois Springfield* Aug 2024 – May 2026

- Managed end-to-end operational workflows for 350+ digital asset remediations (PDF, Word, PowerPoint, video) ensuring compliance with WCAG 2.1, ADA Title II, and Section 508 standards demonstrating regulatory compliance analysis and operational process management.
- Designed and maintained structured workflow tracking and SLA-monitoring systems within the DART queue environment tracking throughput KPIs, turnaround timelines, and service delivery priorities across concurrent high-volume project streams.
- Conducted systematic QA testing using JAWS, NVDA, WAVE, and Accessibility Checker tools applying root-cause analysis methodologies to identify, document, and resolve compliance gaps across diverse content formats.
- Collaborated cross-functionally with faculty, instructional designers, QA reviewers, and university stakeholders to deliver accessible educational content within strict academic deadlines for 500+ students.
- Led workflow bottleneck identification and continuous improvement initiatives standardizing QA documentation procedures and improving operational reporting consistency across the remediation lifecycle.

Data Science Intern | *Coratia Technologies (Teachnook)* Aug 2023 – Sep 2023

- Built automated data processing pipelines using Python and SQL supporting analytical reporting across operational and marketing datasets improving reporting reproducibility and data consistency.
- Conducted exploratory data analysis (EDA), statistical trend evaluation, and customer segmentation on structured datasets to surface business insights and performance optimization opportunities.
- Developed interactive Tableau and Power BI dashboards visualizing KPIs including sales performance, forecasting metrics, and operational trends for stakeholder reporting and executive decision support.
- Applied A/B testing methodologies and predictive analytics to evaluate campaign effectiveness, contributing to data-driven business planning workflows.

Full Stack Development Intern | *SmartInternz EdTech* Jan 2023 – May 2023

- Designed and developed full-stack web applications using React.js, Node.js, MongoDB, and RESTful APIs building structured systems thinking and data flow design skills applicable to enterprise analytics environments.
- Engineered optimized backend database queries and API services, improving data processing efficiency and application performance by 20%.

- Participated in Agile sprint cycles including backlog grooming, QA testing, and collaborative deployment strengthening cross-functional project management skills.

Full Stack Developer Intern AI & Assistive Technology | *Alhansat Solutions with IEEE Bombay Section* Sep 2023 – Oct 2023

- Contributed to AI-driven Text-to-Speech and speech-processing solution development using JavaScript and Node.js supporting accessibility-focused communication platforms.
- Applied NLP-oriented workflows and linguistic analysis techniques to improve speech accuracy and usability across multilingual voice-enabled applications.
- Participated in full SDLC activities including frontend-backend integration, QA testing, debugging, and deployment within collaborative Agile development environments.

PROJECTS

Clinical Operations & Readmission Intelligence Platform | *Healthcare Analytics* | Python · SQL · Tableau 2025

- Analyzed CMS Hospital Readmissions Reduction Program public dataset (3,200+ hospitals) using Python and SQL to identify high-risk patient cohorts, readmission drivers, and operational performance gaps across cardiovascular, pneumonia, and COPD conditions.
- Developed Tableau dashboards tracking operational KPIs including readmission rates by condition, CMS penalty exposure by hospital tier, patient throughput, and volume forecasting applying epidemiological and statistical analysis methodologies.
- Translated complex clinical dataset findings into structured executive narratives and data-driven operational recommendations for hospital planning and healthcare business stakeholders.

Credit Risk Segmentation & Loan Performance Analytics | *Finance Analytics* | Python · SQL · Power BI · Excel Modeling 2025

- Processed and analyzed 2.2M+ loan records from the Lending Club public dataset using Python and SQL performing risk tier segmentation, default rate analysis, and borrower profile classification across income bands, credit grades, and geographic regions.
- Built Power BI dashboards and advanced Excel financial models visualizing default rates, loan grade distributions, geographic risk concentration, and variance analysis supporting credit portfolio monitoring and financial planning workflows.
- Applied statistical modeling and finance/accounting principles to identify top predictors of loan default, enabling risk-informed segmentation strategies aligned with enterprise financial reporting and budgeting environments.

Pharmaceutical Market Analytics & Forecast Intelligence Platform | *Pharma Analytics* | Python · SQL · Power BI · Excel · ARIMA Forecasting 2025

- Integrated CMS Part D Prescriber public data to analyze prescription volume trends, provider-level performance, and regional market dynamics across therapeutic drug categories aligned with pharmaceutical commercial analytics and market research workflows.
- Developed Power BI dashboards and Excel financial models tracking KPIs including prescription growth rates, market share simulation, physician engagement trends, and quarterly revenue forecasting supporting annual budget and periodic forecast deliverables.
- Applied ARIMA-based time series forecasting on prescription volume data to support commercial planning, demand estimation, and operational reporting for pharmaceutical business intelligence environments.

Retail Customer Intelligence & Revenue Analytics Platform | *Business Intelligence* | SQL · Python · Power BI 2025

- Engineered a retail analytics platform using the Olist Brazilian E-Commerce dataset (100K+ orders) to analyze customer purchasing behavior, segment high-value cohorts using RFM methodology, and evaluate revenue performance trends.
- Conducted cohort analysis, churn modeling, and customer lifetime value (CLV) scoring using Python and SQL delivering business insights to optimize marketing strategy, retention planning, and operational sales reporting.
- Built interactive Power BI dashboards visualizing KPIs including CLV, repeat purchase rates, monthly sales growth, and inventory movement supporting data-driven business planning and executive stakeholder reporting.

Intelligent Enterprise Knowledge Assistant (RAG Architecture) | *Solution Design* | Python · LangChain · FAISS · OpenAI GPT · FastAPI · Docker 2025

- Architected an AI-powered Retrieval-Augmented Generation (RAG) system using LangChain, FAISS vector search, and OpenAI GPT designing end-to-end solution workflows for domain-specific enterprise knowledge retrieval and business process automation.
- Designed containerized backend services using Docker and FastAPI, supporting modular and scalable enterprise application integration demonstrating technical specification writing and solution architecture competencies.

- Conducted solution feasibility analysis and workflow documentation, translating business requirements into structured implementation specifications for enterprise-style deployment environments.

Brain Tumor Classification & Diagnostic Analytics System | *Healthcare AI* | *Python · CNN · SVM · PyTorch · OpenCV* 2024

- Developed a hybrid deep learning pipeline combining CNN and SVM models for brain tumor classification on MRI datasets, achieving 99.2% prediction accuracy with reproducible benchmarking and statistical evaluation workflows.
- Applied hyperparameter optimization, feature engineering, and model performance analysis supporting scalable healthcare AI research and clinical diagnostic decision-support applications.

EDUCATION

Master of Science Computer Science | *University of Illinois Springfield (UIS)* Aug 2024 – May 2026
GPA: 3.92 | Relevant: Data Analytics, Machine Learning, Database Systems, Statistical Computing, Software Engineering

Bachelor of Technology Computer Science & Engineering | *SR Guldavaller Engineering College*
GPA: 8.2/10 AP, India 2020 – 2024
Relevant: Data Structures & Algorithms, Database Management Systems, Operating Systems, Web Technologies